

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	29 June 2026
Team ID	
Project Name	Library Book Request System using ServiceNow
Maximum Marks	4 Marks

Technology Stack (Architecture & Stack)

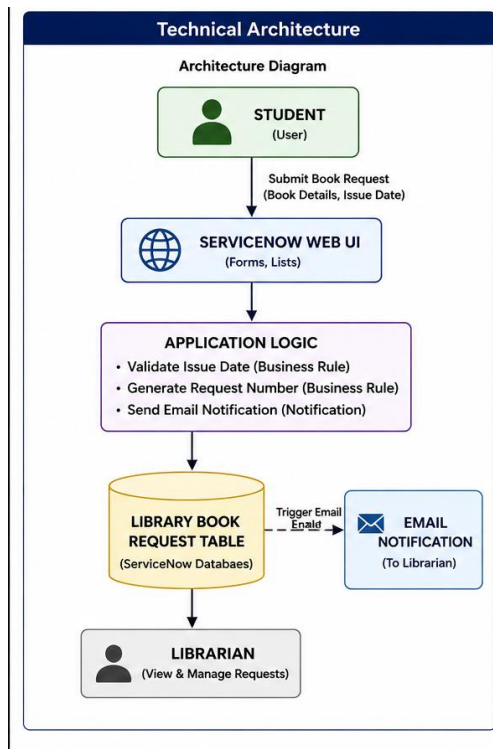


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	User interacts with the application through forms and lists.	ServiceNow Web UI
2	Application Logic-1	Validates that the issue date is not in the past.	ServiceNow Business Rule (JavaScript)
3	Application Logic-2	Automatically generates a unique request number.	ServiceNow Business Rule (JavaScript)
4	Application Logic-3	Sends email notification to the librarian after	ServiceNow Notification

S.No	Component	Description	Technology
		request submission.	
5	Database	Stores library book request records.	ServiceNow Tables
6	Cloud Database	Stores application data in the ServiceNow cloud platform.	ServiceNow Cloud Database
7	File Storage	Stores application records and attachments.	ServiceNow Attachment Storage
8	External API-1	Not Used	N/A
9	External API-2	Not Used	N/A
10	Machine Learning Model	Not Used	N/A
11	Infrastructure (Server / Cloud)	Application hosted on the ServiceNow cloud platform.	ServiceNow Cloud

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No external open-source framework is used.	ServiceNow Platform
2	Security Implementations	Role-based access control and authenticated user access.	ServiceNow ACL & User Authentication
3	Scalable Architecture	Cloud-based architecture supports increasing numbers of requests.	ServiceNow Cloud Platform
4	Availability	Application is available online through the ServiceNow platform.	ServiceNow SaaS
5	Performance	Business Rules and Notifications provide efficient request processing.	ServiceNow Platform